

Brian Lee

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[LinkedIn](#) | [Github](#) | [Personal Website](#)

Experience

Software Develop iXp | SAP

Nov 2025 – Ongoing

- Developed reusable React/TypeScript component architecture adopted across 5+ enterprise product features, improving development consistency
- Integrated React frontend with SAP OData services and SpringBoot APIs
- Built automated component and end-to-end testing suites using Cypress and Cucumber, improving release reliability and reducing regression issues

Software Developer Co-op | Caracal Technologies Inc.

Jan 2024 – Aug 2024

- Redeveloped the reporting interface with React and TypeScript, improving performance and refactoring the codebase into a maintainable component-based structure
- Utilized Selenium to automate processing time from 8 hours to minutes
- Engineered a secure file explorer using AWS S3 and pre-signed URLs, featuring embedded previews and automated CI/CD deployment via CloudFront

Projects

Potential | [Github](#) | [Demo](#)

April 2026 – Ongoing

- Engineered a location intelligence platform using FastAPI and PostGIS that predicts business favorability for any Vancouver coordinate using ML and open geospatial data
- Trained category-specific XGBoost machine learning model on Canada census demographics and OSM business density, achieving an R^2 of 0.58
- Designed high-efficiency PostGIS queries to perform geospatial joins across 300+ census tracts, mapping real-time coordinate clicks to localized demographic insights
- Developed a type-safe backend using SQLAlchemy and Pydantic, managing database schema evolutions with Alembic migrations
- Deployed containerized FastAPI backend via Docker with Neon PostgreSQL, and Next.js/Shadcn/UI frontend on Vercel with MapLibre GL map visualization

Technical Skills

Languages & Frontend: Python, TypeScript/JS, Java, R, React, Next.js, MapLibre GL, Tailwind, Selenium

Backend & Data: FastAPI, Spring Boot, Express, PostgreSQL (PostGIS), SQLAlchemy, Alembic, Docker, AWS (S3/Lambda)

AI & Tools: XGBoost, Scikit-learn, Pandas, Pydantic, Git, Postman, Jira, Agile

Education

British Columbia Institute of Technology – Computer Science

Jan 2023 – May 2025

University of Toronto - Bachelor of Science - Major in Mathematics

Sep 2013 – May 2018